CS760 Midterm Exam

Instructions: Answer the following questions. Write out solutions below each question. Even if you cannot obtain a final answer, make sure to write your setup and explain how you would obtain the answer. Partial credit will be considered. Do not spend too much time on any single problem, as some problems are harder than others. Try to write only as much as needed to answer the question; the short answer questions that do not require derivations can typically be answered in 1-2 short sentences.

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You get 3 bonus points for writing your name and Wisc ID correctly. (3 pts)

1 Evaluation

1.1 Metrics [5 points]

5

Describe one binary classification setting where accuracy might not be the best metric, and argue in favor of another metric instead.

Binary classification setting: is it raining outside — yes(1) / no(0)

It can be easy to train because it does not need a lot of parameters.

Ex: is it cloudy — yes ⇒ rain / no ⇒ no rain.

We also do not need huge data to train it.

1.2 Validation [5 points]

0

Suppose I split my data into a training, validation, and test split, and I try to do empirical risk minimization with gradient descent. Which split do I use in each of the following scenarios, and why?

1. Tuning the number of iterations of gradient descent to obtain a better generalization error.

Here I would use k-fold cross validation. It best appoximates tre success against unseen data.

2. I want to check if gradient descent is actually minimizing the empirical risk.

Here I would use stratified sampling, because that way my datasets will have Proportional instances.

3. I want to release my model along with an estimate of how it performs on unseen data.

Here I would use random splits, since that would be the best way to approximate unseen data.

loso si razbral prasanjeto:ne se odnesuva na toa kako delis primerok. tuku na koj od setovite da izberes vo dadenata situacija

22. ReLU =>

ReLU(x) = max(0, x)

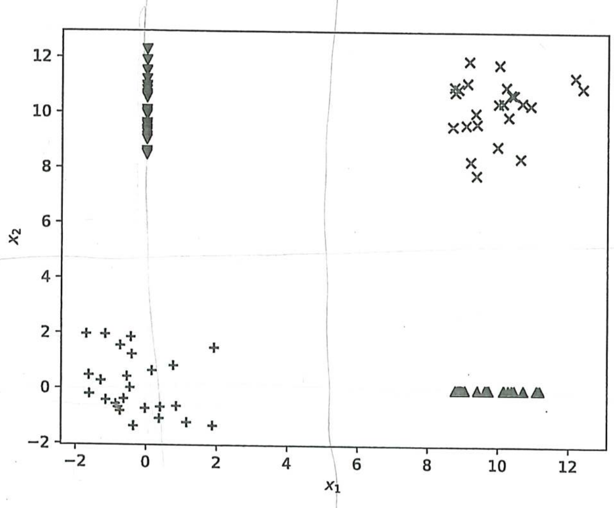


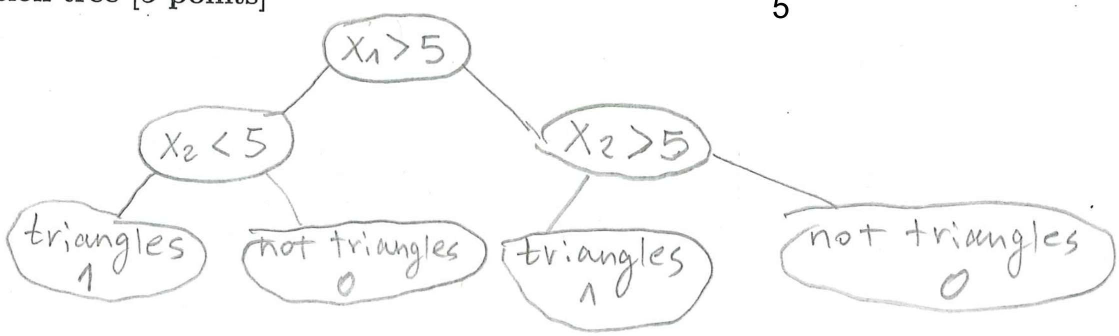
Figure 1: Data for Question 2 (and 3)

2 Supervised modeling

For each model class below, describe a simple instance that can perfectly classify the points in Figure 1 into "triangles" ($y = 1$) or "not triangles" ($y = 0$), or say briefly why it cannot be done. For the parametric models, you can use a hard threshold ($h(z) = 1_{z \geq 0}$) as the last layer rather than a sigmoid.

2.1 Decision tree [5 points]

5



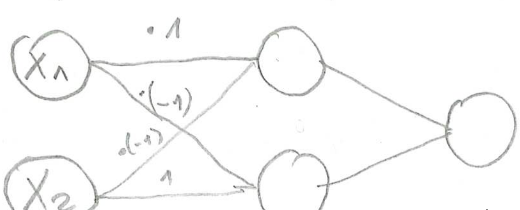
2.2 Linear classifier [5 points]

This cannot be done because there exists no line (only one line) that can separate the data, so that on one side we have triangles, and on the other side non triangles.

2.3 2-layer MLP with ReLU activation [5 points]

5

h_1 = ReLU(wX)



x∈{

intuition: when X_1 and X_2 are both 0. or 10 => we do not have (▽, △)

When only 1 is 0 and the other one is 0 => we have triangles (△, ▷)

In case the MLP does not do that.

nemas tezini na drugi dva rebra. -2

inaku ova ti e XOR sto go gledavme

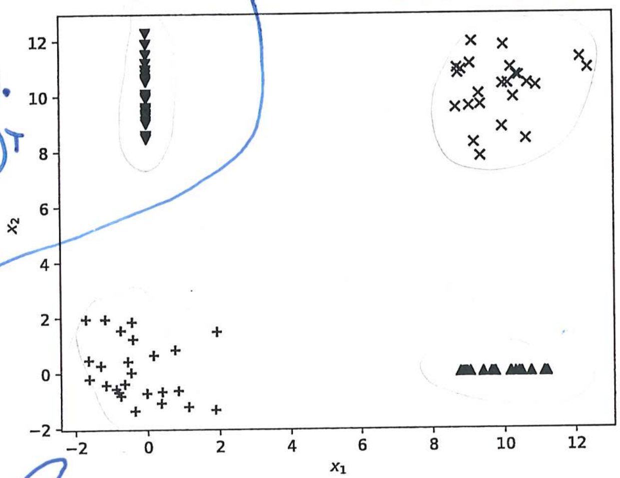
3.1. $\phi_j = \frac{1}{n} \sum_{i=1}^{n} w_j^{(i)}$

$\mu_j = \frac{\sum_{i=1}^{n} w_j^{(i)} \cdot x^{(i)}}{\sum_{i=1}^{n} w_j^{(i)}}$

$\Sigma_j = \frac{\sum_{i=1}^{n} w_j^{(i)} (x^{(i)} - \mu_j^{(i)}) \cdot (x^{(i)} - \mu_j^{(i)})^T}{\sum_{i=1}^{n} w_j^{(i)}}$

In case my approx for Σ aren't correct:
Σ determines the variances of the data type

$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$ ~~For x and + ... ~ 1.~~
For Δ and ∇ ~~main diagonal ~ 1~~
~~Second diagonal = 0~~
~~because there is no variance in one of the dimensions~~
(for Δ $\sigma_{x_2} = 0$
for ∇ $\sigma_{x_1} = 0$)



Main diagonal: ⬭ or ⬯

Second diagonal: ⬯ or ⬯.

Figure 2: Data for Question 3 (and 2)

$1 \Rightarrow +$
$2 \Rightarrow \times$
$3 \Rightarrow \Delta$
$4 \Rightarrow \nabla$

3 Unsupervised modeling

Below we refer to the points in Figure 2 (note this is the same figure as Figure 1).

3.1 Gaussian mixture model [5 points]

Suppose the data was generate by a Gaussian mixture model with four latent variables. Write down parameters $\phi_1, \ldots, \phi_4 \in [0, 1]$, $\mu_1, \ldots, \mu_4 \in \mathbb{R}^2$ and $\Sigma_1, \ldots, \Sigma_4 \in \mathbb{R}^{4\times 4}$ (2×2) of a four-component Gaussian mixture model that could plausibly (in a probabilistic sense) generate the data. For simplicity, make the first latent variable (cluster) correspond to +, the second to x, the third to the up-pointing triangle ($\triangle$), and the fourth to the down-facing triangle. For simplicity, make the probability of each cluster be the same (so you do not need to count the number of points in each cluster) and only use the numbers 0, 0.25, 1, and 10.

$\mu_1 = [0, 0]^T$
$\mu_2 = [10, 10]^T$
$\mu_3 = [10, 0]^T$
$\mu_4 = [0, 10]^T$

$\Sigma_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$
$\Sigma_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$
$\Sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
$\Sigma_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

$\phi_1 = 0$
$\phi_2 = 0$
$\phi_3 = 1$
$\phi_4 = 1$

receno e deka probabilities are the same
f1=0.25,

3.2 Principal components analysis [5 points]

List one reason why principal components analysis (PCA) might not be useful to apply to this data.

Pca is useful for dimensionality reduction. We need a data that has high variance in one or more dimensions. Here our data is relatively equally distributed. There is not one dimension that dominates the other.

4.3. $\theta_{t+1} = \theta_t - \eta \cdot \nabla_\theta L(\theta_t; x_i, y_i)$

-eray
formula

4 Optimization

Convex => it has to have a gradient that finds the MLE.

In this question, we will consider optimization of the following objective:

$$f(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} f_i(\mathbf{x})$$

We assume each $f_i : \mathbb{R}^d \mapsto \mathbb{R}$ is convex and has L-Lipschitz gradients and is convex, for known $L > 0$.

4.1 Gradient descent (GD) update [5 points]

dosta teoretsko prasanje,

Given we are at some point $\mathbf{x}_{t-1} \in \mathbb{R}^d$, write down the tth iteration of GD given a fixed step-size $\eta > 0$ and state how many gradient computations are required to compute it.

$X_{t-1} = X_t - \eta \nabla$

ti dadoa ocenka samo za formula, prasanjeto e malku triki, bidejki
imas n funkcii fi(x) znaci ti trebaat n gradienti (za sekoja fi(x)

4.2 Number of iterations for gradient descent [5 points]

As we saw in class, if $\eta \leq 1/L$ then GD on f has suboptimality $f(\mathbf{x}_T) - f(\mathbf{x}^*) \leq \frac{\|\mathbf{x}_0 - \mathbf{x}^*\|_2^2}{2\eta T}$ after T iterations. Show that this implies the existence of a step-size that guarantees ε-suboptimality, i.e. reaching a point $\mathbf{x}_T \in \mathbb{R}^d$ s.t. $f(\mathbf{x}_T) - f(\mathbf{x}^*) \leq \varepsilon$, if we run gradient descent for at least $T = \frac{L\|\mathbf{x}_0 - \mathbf{x}^*\|_2^2}{2\varepsilon}$ iterations.

Num of iters = 1.

Num of iterations = num of data points.

4.3 SGD update [5 points]

Given we are at some point $\mathbf{x}_{t-1} \in \mathbb{R}^d$, write down the tth iteration of SGD given a fixed step-size $\eta > 0$ and state how many gradient computations are required to compute it.

barem ova mozese da odgovoris, SGD zamenuva suma so samo edna fi(x)

4.4 Comparing bounds [5 points]

SGD on f is guaranteed to yield a point with (expected) suboptimality $\leq \varepsilon$ after $T = \left(\frac{2L\|\mathbf{x}_0 - \mathbf{x}^*\|_2^2 + C/L}{\varepsilon}\right)^2$ iterations, for constant $C \geq 0$ (Garrigos & Gower, Corollary 5.6). At least according to this upper bound and the one from Question 4.2, and ignoring expectations, show that we can have at most $n = \mathcal{O}(1/\varepsilon)$ functions f_i in the summation $f = \sum_{i=1}^{n} f_i$ for the gradient descent guarantee to be *better* than the SGD guarantee, in terms of the **number of gradient computations** sufficient to attain a fixed suboptimality $\varepsilon > 0$.

4.5 When to use SGD instead of gradient descent [5 points]

If each i corresponds to a data point with loss f_i and minimizing $f = \sum_{i=1}^{n} f_i$ is thus doing empirical risk minimization, what does the previous bound suggest about the number of data points n and the required optimization precision (suboptimality) ε where it is often better to run SGD than gradient descent?

i ova eventuelno diskutiravme ova

5 Regression

Suppose we model the data-generation process as $y = \mathbf{w}^\top \mathbf{x} + \epsilon$, where inputs $\mathbf{x} \in \mathbb{R}^d$ and an unknown parameter $\mathbf{w}$ are d-dimensional and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ for some $\sigma \geq 0$. Recall that the p.d.f. of a normally distributed random variable $Y \sim \mathcal{N}(\mu, \sigma^2)$ is $f_Y(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)$.

5.1 Conditional probability [5 points]

Write down an expression for $P(y|\mathbf{x}, \mathbf{w})$.

$$P(y|x,w) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-wx)^2}{2\sigma^2}\right)$$

5.2 Conditional log-likelihood [5 points]

Assume you are given a set of training observations $(\mathbf{x}^{(i)}, y^{(i)})$ for $i = 1, \ldots, n$. Derive the conditional log-likelihood $\log P(\mathbf{y}|\mathbf{X}, \mathbf{w})$ of this training data, where $\mathbf{y} \in \mathbb{R}^n$ and $\mathbf{X} \in \mathbb{R}^{n \times d}$ aggregate the labels $y^{(i)} \in \mathbb{R}$ and inputs $\mathbf{x}^{(i)} \in \mathbb{R}^d$ into a vector and a matrix, respectively. Do NOT drop any constants.

→ here x is not a matrix.

$$\log f_n(\mu,\sigma^2) = \ell(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y^{(i)} - wx_i^{(i)})^2$$

$$\log(f_n(\mu,\sigma^2)) = \ell(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}(y^{(i)} - wX)^2$$

$\ell(w)$

5.3 Equivalence to ordinary least-squares [5 points]

Argue briefly why maximizing the log-likelihood over $\mathbf{w}$ yields the same solution as minimizing $\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$.

Since (both) elements, especially the element including w is negative, and we are trying to find the max, we need to find the smallest of the absolute value of that element. Second explanation: by mathematics/Algebry $\max(-x) = \min(x)$

5.4 Deriving the MLE [5 points]

Assuming $\mathbf{X}$ has full column rank, show that the MLE is $(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$.

Since it is not $n \times n$ matrix we cannot use ~~[illegible]~~. $w = X^{-1}y$.

$w_{MLE} = \arg\max_w P(X|w)$; ~~loglikelihood~~ the derivative of $\ell = 0$.

$$\frac{\partial}{\partial w}\ell(w) = \left(\cancel{-\frac{n}{2}\ln(2\pi\sigma^2)} - \frac{1}{2\sigma^2}(y^{(i)} - wX)^2\right)' = 0$$

no w

$$-\frac{1}{\cancel{2}\sigma^2} \cdot \cancel{2}(y^{(i)} - wX)\cdot(-1\cancel{X}) = 0 \quad \Rightarrow \frac{X}{\sigma^2}(y^{(i)} - wX) = 0$$

$$w = \frac{Xy}{\sigma^2}$$

$$\frac{Xy}{} - wX^2 = 0 \qquad w = \frac{Xy}{}$$

ne gledam deka si ja izvel formlata do kraj. Trebase da dobies
dadenata formula. Baska ti resavas kako da ova ne se matrici

$$L(w) = (y - Xw)^T (y - Xw) + \lambda \|w\|_1$$

5.5 Bayesian view of LASSO [5 points]

Assume a prior on $\mathbf{w}$ that each of its d entries is drawn i.i.d. from a Laplacian distribution $\text{Lap}(0, b)$. Write down the posterior probability $P(\mathbf{w}|\mathbf{y}, \mathbf{X})$ of $\mathbf{w}$, dropping scaling terms that do not depend on $\mathbf{w}$ (i.e. you can just derive what the posterior is proportional to). Recall that the p.d.f. of a Laplace-distributed random variable $Z \sim \text{Lap}(\mu, b)$ is $f_Z(z) = \frac{1}{2b} \exp\left(-\frac{|z-\mu|}{b}\right)$. Hint: use Bayes' rule, i.e. $P(A|B) = P(B|A)P(A)/P(B)$ for random variables A and B satisfying $P(B) \neq 0$.

5.6 MAP estimation [5 points]

Show that the maximum *a posteriori* estimator of $\mathbf{w}$ with this prior is equivalent to minimizing the LASSO objective, which is $\frac{1}{2}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda\|\mathbf{w}\|_1$ for $\lambda = \frac{\sigma^2}{b}$.

$$W^{MAP} = \underset{w}{\text{argmax}} \prod_{i=1}^{n} P(X | w) \cdot p(w)$$

tuka ti dade 2 poena samo za formula za MAP
ova imase za HW
jas ti go izvedov celoto i ti rekov da dopravis vo domasnoto

5.7 Effect of the prior on regularization strength [5 points]

How does varying the scale parameter b of the Laplace prior affect the regularization strength λ, and why does this effect make sense from the Bayesian view in which the entries of $\mathbf{w}$ being sampled from $\text{Lap}(0, b)$?

Prior is exactly the strenght, shape and bias of that extra term we add to the loss, and its gradient.
Smaller b, larger λ. $\Rightarrow$ stronger regularization and more shrinkage.

5.8 LASSO vs. Ridge regression [5 points]

State a property that an estimate of $\mathbf{w}$ computed via LASSO typically have that Ridge typically does not, and list a reason why this property can be useful.